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A formulation to optimize the risk reduction process based on LOPA

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ABSTRACT

This paper presents a mixed integer nonlinear programming (MINLP) model to improve the computational use of the layer of protection analysis (LOPA). For a given set of independent protection layers to be implemented in a process, the proposed optimization model is solved to: a) Include costs associated with the different prevention, protection and mitigation devices, and b) Satisfy the risk level typically specified in the LOPA analysis through the occurrence probability. The underline purpose focuses on improving the analysis process and decision making to obtain the optimal solution in the safeguards selection that satisfies the requirements to be considered as IPL's. The optimization is based on economic and risk tolerance criteria. As a first stage of this proposal, the safety instrumented system (SIS) design is optimized so that the selection of SIS components minimizes the risk and satisfies the safety integrity level (SIL) requirements. A case study is presented to validate the whole proposed approach.

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1. Introduction

All chemical plants handle, process and storage large amounts of hazardous materials. Therefore, several tasks must be carried out to produce a safety environment for their employees. Risk analysis methodologies have been developed to meet these requirements. These risk analysis techniques have been classified in four categories according to particular properties: Deterministic, probabilistic, qualitative and quantitative (Tixier, Dusserre, Salvi, & Gaston, 2002). Some of the important risk analysis methods used in any of these categories are: Hazard and Operability (HAZOP), the qualitative "What if?" analysis, Event Tree Analysis (ETA), Fault Tree Analysis (FTA), Failure Mode Effect Criticality Analysis (FMECA), Quantitative Risk Assessment (QRA) and layer of protection analysis (LOPA).

LOPA has been broadly accepted as a risk analysis and assessment tool and several works have been published in the literature where it has been successfully applied. The safety integrity level (SIL) for a safety instrumented system (SIS) has been determined through applying LOPA (Dowell, 1998, 1999). The LOPA approach has been used to estimate reactive chemical risk based on frequencies of failure scenarios for the hydroxylamine production system (Wei, Rogers, & Mannan, 2008). Markowski and Mannan (2009) explored the application of fuzzy logic for risk assessment of major hazards connected to transportation of flammable substances in long pipelines. Their basis for risk assessment was the fuzzy layer of protection analysis (fLOPA) framework. A Bayesian-LOPA estimation that can update general failure with plant-specific failure data has been used to estimate potential risks on LNG importation terminals (Yun, Rogers, & Mannan, 2009). More recently, Markowski and Mannan (2010) have introduced the concept of ExSys–LOPA which combines expert system ExSys and LOPA methodology to assess risk.

When optimal solutions become a part of the main goals, it is often important to provide a cost-benefit analysis. However, there is a lack of works in the literature to provide a convenient risk cost analysis of the protective measures. Pasman (2000) has proposed an optimization model to include safety costs such as costs of the accident, accident frequency and costs of risk reduction measures. Four options for risk reduction measures have been considered: No measure, small, medium and large. Latter, LOPA was incorporated into a design support system to produce a cost-effective approach during the earliest conceptual design (Schupp, Hale, Pasman, Lemkovitz, & Goossens, 2006). However, an optimal allocation of protective devices has not been sufficiently dealt in published works. More recently, Stauffer (2010) describes a procedure to select the best alarms system adjoined to an appropriate operators training to reduce the SIL required for the SIS.

This research aims to produce an approach to optimize each layer of LOPA for process systems to reduce risk. This work presents

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an optimization approach that includes each independent layer of protection with its associated costs and probability of failure on demand into a MINLP formulation. Solving the model will provide the optimal allocation of independent protection layer to take decisions based on minimizing the costs but also satisfying the risk tolerance criteria. This approach focuses on detailing the SIS layer though the overall idea is applicable on all the protective layers. Next section provides a brief description of LOPA.

2. Layer of protection analysis (LOPA)

LOPA is a semi-quantitative methodology used in risk analysis. LOPA assesses the scenarios such as loss of containment, which can lead into major accidents, and it also proposes a series of hierarchically organized protective layers. Protective layers are placed to lower the frequency of undesired consequences. The methodology typically uses order of magnitude to express the initial event frequency, the probability of failure on demand of the independent protection layers and the magnitude of the consequence.

LOPA methodology typically builds on the information developed during a qualitative hazard evaluation. Then, layers of protection are intended to independently comply with three main functions: Prevention, protection and mitigation. To be considered as independent protection layers (IPL's), safeguards need to satisfy some characteristics: independence, specificity, dependability and auditability (Summers, 2003).

LOPA methodology typically uses order of magnitude for initiating events frequency, consequence severity and the likelihood of failure of IPL's to evaluate a risk scenario. The first step consists in defining a risk scenario described by the initial event–consequence pair. Then the frequency of the initial event is estimated. Furthermore, it is necessary to identify each IPL to evaluate its probability of failure on demand, estimate the frequency of the consequence, and estimate risk as the product of frequency with estimated severity. Finally, the risk is evaluated and compared to the estimated risk until the risk tolerance criteria is satisfied.

The first layer of protection is the process design where inherently safer process should be preferred. The second layer of defense corresponds to applying basic process control systems followed by process alarms and human intervention. Next layer of protection is formed by the SIS, which are placed to increase a reduction in the probability of hazardous situation occurrences caused mainly by human errors. SIS layer is then hierarchically followed by physical protections such as emergency pressure reliefs. A further protection layer is the installation of fire and blast walls or dikes, and the final layers refer to plant emergency and community response. The following section describes relevant factors in the SIS design.

3. Safety instrumented systems (SIS)

SIS is an independent protection layer installed to reduce the risk associated with a hazardous scenario. A SIS is composed of sensors, logic solvers and final elements or actuating items. SIS implement safety instrumented function (SIF) intended to archive or maintain a safe state with respect to a specific process demand. The basic principles for SIS design are described in the IEC 61508 norm (Smith & Simpson, 2011). The main purpose of this norm is to ensure that the safety level given by the SIL is effectively achieved based on the functionality of all electric and electronic devices. The norm IEC 61508 is not a specific document for a particular industry but a generic one, which has been applied in several sectors such as aircraft industry, process industry, nuclear, automotive, marine, etc.

The following necessary factors in SIS design are indicated in the IEC 61508 norm: Specific architecture (redundancy scheme), hardware types: A or B, multiple failure modes (safe and dangerous failure rates), diagnostic coverage (share of detected and undetected failures), common cause failure (β factor), proof test intervals, and repair times. The norm also includes two indicators: The fault tolerance (FT) that is the number of faults that a subsystem can tolerate before resulting in system failure, and the safe failure fraction (SFF) that is the percentage of subsystem failures considered safe. Table 1 shows the relation between these two indicators.

The probability of failure on demand (PFD) for each SIS device is required for the risk evaluation. The average PFD of a safety function for a safety instrumented system is determined by calculating the average probability of failure on demand for all the subsystems and combining all of them to provide the safety function. Several methods have been proposed to perform this evaluation: Simplified equations (SE), reliability block diagrams (RBD), fault tree analysis (FTA), Markov analysis (MA), Petri nets, and hybrid methods. Among them, RDB, FTA and MA have been preferred though a debate remains about what it could be the best choice. Several authors prefer FTA because it is powerful for easy modeling of large and complex systems, and their results are very good when small probabilities are involved such as the typical values in safety systems (Andrews & Ericson, 2000; Goble & Cheddie, 2005; Torres-Echeverria, 2009a, 2009b). In this work, the FTA is used to calculate all PFD's. The following section describes the mathematical model developed to optimize LOPA.

4. Mathematical formulation

The total cost of the implemented protection layers, C^{T}, contains the sum of fixed, operation and risk costs. The total cost is given by:

$$C^{T} = C_{T}^{Fixed} + \left(C_{T}^{Op} + C^{Risk}\right)PV_{A} \tag{1}$$

where C_{T}^{Fixed} corresponds to the total initial fixed costs of all independent protection layer implemented, C_{T}^{Op} is the total operation cost of all de allocated independent protection layers, C^{RISK} is the risk cost, and PV_{A} is the present value of an annuity factor. These costs are estimated by:

$$C_{T}^{Fixed} = \sum_{L} C_{L}^{Fixed} \tag{2}$$

$$C_{T}^{Op} = \sum_{L} C_{L}^{Op} \tag{3}$$

$$PV_{A} = \frac{1-(1-R)^{-t}}{R} \tag{4}$$

where R represents the discount rate, which combines the interest rate and the inflation rate (Goble, 2010). The term t represents the lifecycle of the project in years.

The risk cost per year is the sum of the costs generated by spurious trips, C^{STR}, and the expected loss due to a dangerous failure, C^{HAZARD}, as detailed in (Torres-Echeverria, 2009a, 2009b):

Table 1
SIL Architectural constraints (IEC 61508, 1998–2005).

Subsystem type	A			B		
Hardware FT	0	1	2	0	1	2
SFF						
<60%	SIL 1	SIL 2	SIL 3	N/A[a]	SIL 1	SIL 2
60–<90%	SIL 2	SIL 3	SIL 4	SIL 1	SIL 2	SIL 3
90–<99%	SIL 3	SIL 4	SIL 4	SIL 2	SIL 3	SIL 4
>99%	SIL 3	SIL 4	SIL 4	SIL 3	SIL 4	SIL 4

[a] Not allowed.

$$C^{\mathrm{RISK}} = C^{\mathrm{STR}} + C^{\mathrm{HAZARD}} \tag{5}$$

The C^{STR} is not generated for all IPL's but only for the active barriers where the occurrence of a safe failure can lead to loss production. The cost of spurious trip rate C^{STR} is given by:

$$C^{\mathrm{STR}} = \mathrm{STR}\cdot\mathrm{CSD} \tag{6}$$

where STR is the spurious trip rate and CSD is the spurious shutdown cost. The CSD is given by:

$$\mathrm{CSD} = \mathrm{SD}_{\mathrm{Loss}}\mathrm{SD}_{\mathrm{Time}} \tag{7}$$

where $\mathrm{SD}_{\mathrm{Time}}$ is the loss production time d and $\mathrm{SD}_{\mathrm{Loss}}$ is the production loss.

The cost of hazard C^{HAZARD} is given by:

$$C^{\mathrm{HAZARD}} = C^{\mathrm{C}}\cdot f^{\mathrm{I}}\cdot \mathrm{PFD}^{\mathrm{NON\text{-}SIS}}\cdot \mathrm{PFD}^{\mathrm{SIS}} \tag{8}$$

where C^{C} is the economic catastrophic loss, f^{I} is the initial event frequency, $\mathrm{PFD}^{\mathrm{NON\text{-}SIS}}$ is the probability of failure on demand of non-SIS independent protection layers and $\mathrm{PFD}^{\mathrm{SIS}}$ is the probability of failure on demand for SIS.

In order to ensure the fulfillment of safety criteria, the optimization is subject to the constraint given by:

$$f^{\mathrm{TR}} \geq f^{\mathrm{I}}\mathrm{PFD}^{\mathrm{NON\text{-}SIS}}\mathrm{PFD}^{\mathrm{SIS}} \tag{9}$$

For the SIS, the lifecycle costs are estimated with the following equations:

$$C^{\mathrm{SIS}} = C^{\mathrm{Fixed}} + \left(C^{\mathrm{Op}} + C^{\mathrm{Risk}}\right)\cdot \mathrm{PV}_{\mathrm{A}} \tag{10}$$

where initial fixed costs C^{Fixed} include the design, purchase, installation, commissioning costs C_j^{DPIC} per j-subsystem and start-up costs $C^{\mathrm{START\text{-}UP}}$:

$$C^{\mathrm{Fixed}} = \sum_j C_j^{\mathrm{DPIC}} + C^{\mathrm{START\text{-}UP}} \tag{11}$$

Since a subsystem could include sensors and logic solvers or final elements in the safety instrumented system, then the overall operating costs per year includes testing costs, C^{T}, preventive maintenance costs, C^{PM}, and corrective maintenance costs, C^{CM}.

$$C^{\mathrm{Op}} = C^{\mathrm{T}} + C^{\mathrm{PM}} + C^{\mathrm{CM}} \tag{12}$$

Testing costs are defined by:

$$C_j^{\mathrm{T}} = \sum_i \sum_r \frac{1}{\mathrm{TI}} C_{ij}^{\mathrm{T}} N_r B_{ijr} \tag{13}$$

where: TI is the test interval, C_{ij}^{CM} is the cost of testing, N_r is the number of redundancies, B_{ijr} is the binary variable that corresponds to the number of r redundancies selected, with the i-type of technology of the j-subsystem.

Preventive maintenance costs are defined by:

$$C_j^{\mathrm{PM}} = \sum_i \sum_r \frac{1}{M_{ij}} C_{ij}^{\mathrm{PM}} N_r B_{ijr} \tag{14}$$

where: M_{ij} is the maintenance frequency, C_{ij}^{PM} are the preventive maintenance costs for the i-type of technology for the j-subsystem, and N_r is the number of redundancies.

Corrective maintenance costs are calculated by:

$$C_j^{\mathrm{CM}} = 8760 \sum_i \sum_r f_{ij} C_{ij}^{\mathrm{CM}} N_r B_{ijr} \tag{15}$$

where f_{ij} is the frequency of repair given by $f_{ij} = \lambda_{ij}^{\mathrm{T}}$, and $\lambda_{ij}^{\mathrm{T}}$ is the total failure rate that includes safe and dangerous failures into detected and undetected failure modes.

$\mathrm{PFD}_{\mathrm{avg}}$ and STR are quantified when a fault tree analysis is performed. The main structures in safety systems are series, parallel and MooN structures. In this work, the analyzed systems have a MooN structure with $M = 1$. For the estimation of the $\mathrm{PFD}_{\mathrm{avg}}$, it is assumed that the system operates in a low-demand mode and the failure rate is constant.

The average probability of failure on demand for a safety function in the SIS is determined by combining the average probability of failure on demand for all the j-subsystems:

$$\mathrm{PFD}_{\mathrm{avg}}^{\mathrm{SIS}} = \sum_j \mathrm{PFD}_j \tag{16}$$

The proposed model incorporates redundancy allocation with diverse technology options.

Diversity is introduced as a defense against a common cause failure (CCF). It has a significant influence on system's dependability. To select the configuration for the subsystem j having the redundancy r for the i-technology, the following equation is used:

$$\mathrm{PFD}_j = \sum_i \sum_r \mathrm{PFD}_{ijr} B_{ijr} \tag{17}$$

For redundancies higher than two CCF affecting all redundant components, all $\mathrm{PFD}_{\mathrm{avg}}$ for each j-subsystem without diverse technologies can be estimated with the following equation:

$$\begin{aligned}\mathrm{PFD}_{ijr} =& \frac{\left[\lambda_{ij}^{\mathrm{DU}}\left(1-C_{\mathrm{MooN}}\beta_{ij}^{\mathrm{DU}}\right)(8760\cdot\mathrm{TI}+\mathrm{Tr})+\lambda_{ij}^{\mathrm{DD}}\left(1-C_{\mathrm{MooN}}\beta_{ij}^{\mathrm{DD}}\right)\mathrm{Tr}\right]^{N_{ijr}+1}}{8760\cdot\mathrm{TI}\lambda_{ij}^{\mathrm{DU}}(N_{ijr}+1)\left(1-C_{\mathrm{MooN}}\beta_{ij}^{\mathrm{DU}}\right)} \\ &- \frac{\left[\lambda_{ij}^{\mathrm{DU}}\left(1-C_{\mathrm{MooN}}\beta_{ij}^{\mathrm{DU}}\right)\mathrm{Tr}+\lambda_{ij}^{\mathrm{DD}}\left(1-C_{\mathrm{MooN}}\beta_{ij}^{\mathrm{DD}}\right)\mathrm{Tr}\right]^{N_{ijr}+1}}{8760\mathrm{TI}\lambda_{ij}^{\mathrm{DU}}(N_{ijr}+1)\left(1-C_{\mathrm{MooN}}\beta_{ij}^{\mathrm{DU}}\right)} \\ &+\lambda_{ij}^{\mathrm{DU}}C_{\mathrm{MooN}}\beta_{ij}^{\mathrm{DU}}(4380\cdot\mathrm{TI}+\mathrm{Tr})+\lambda_{ij}^{\mathrm{DD}}C_{\mathrm{MooN}}\beta_{ij}^{\mathrm{DD}}\mathrm{Tr}\end{aligned} \tag{18}$$

where $\lambda_{ij}^{\mathrm{DU}}, \lambda^{\mathrm{DD}}$ are the failure rate for dangerous undetected and dangerous detected failures, respectively, C_{MooN} is the configuration factor β^{DU} and β^{DD} are parameters corresponding to the β factor model that is a basic estimation of Common Cause Failures, TI is the test interval, Tr is the repair time, and N_{ijr} is the number or r redundancies.

The spurious trip rate is estimated for subsystems with redundant components and no diversity with the following equation:

$$\mathrm{STR} = \left[\lambda_{ij}^{\mathrm{SU}}\left(1 - C_{\mathrm{MooN}}\beta_{ij}^{\mathrm{SU}}\right)N_{ijr} + \lambda_{ij}^{\mathrm{SD}}\left(1 - C_{\mathrm{MooN}}\beta_{ij}^{\mathrm{SD}}\right)\right]8760 \tag{19}$$

where $\lambda_{ij}^{\mathrm{SU}}, \lambda^{\mathrm{SD}}$ are the failure rate for safe undetected and safe detected failures, respectively.

The $\mathrm{PFD}_{\mathrm{avg}}$ is calculated when two technologies, A and B, are selected. For simplicity, it has been assumed that appropriate measures can eliminate coupling among different technologies so that common cause failures of different technologies are negligible. The PDF and STR should be modified when more technologies appear and the FTA is modified accordingly. The PFD is estimated by:

$$PFD_{avg}^{AB} = \lambda_A^{DU}\lambda_B^{DU}\left(\frac{TI^2}{3} + TI\cdot Tr + Tr^2\right) + \left(\lambda_A^{DU}\lambda_B^{DD} + \lambda_A^{DD}\lambda_B^{DU}\right) \times \left(\frac{TI}{2} + Tr\right) + \lambda_A^{DD}\lambda_B^{DD} \quad (20)$$

The spurious trip rate is,

$$STR = \lambda_A^{SU} + \lambda_A^{SD} + \lambda_B^{SU} + \lambda_B^{SD} \quad (21)$$

5. Case study

The production of hydroxylamine (HA) is important in several sectors of the chemical industry. The reaction is highly exothermic and it tends to increase the pressure in the container:

$$(NH_2OH)_2H_2SO_4 + 2KOH \Rightarrow 2NH_2OH + K_2SO_4 + 2H_2O$$

Whenever the process becomes out of control, some containers may produce a violent explosion. At least two incidents have been recently analyzed since they produced at least nine fatal results (Wei et al., 2008). The model developed before is applied to the HA process and the process flowsheet as well as probabilities of initial events, and IPL's are also taken from published results (Wei et al., 2008) where LOPA has been applied.

The model described before can include several devices but the cost and PDF are required for each device. In this case, the rupture of the HA reactor is the only failure analyzed. This incident is produced by a failure in the so called KOH-control that is used to avoid the thermal decomposition of HA. A runaway reaction is produced under a failure rate of one in ten years (10^{-1}/year). The initial event occurs in the BCPS and, hence, only the IPLs and subsequent layers are used in LOPA to reduce the risk. The layer corresponds to the human participation followed by a SIS and releasing devices.

To apply our approach, it is necessary to include all costs associated to each PFD in each protection layer. Unfortunately, this is not an easy task since it is extremely complex to associate a cost when human being is involved. According to the experts, it is not possible to associate training time to some typed of failures such as stress, working hours, etc. but these factors may considerably affect operators' performance. Hence, a constant value 10^{-1} is used here as indicated in (CCPS, 2001), and $70,000 usd as its associated cost. The SIS layer allows testing several subsystems, sensors, logic systems and final elements as well as their combinations. It then allows the possibility of exploring redundancies in both devices and technologies. For the case study, five technologies are used in each subsystem. Thus, data for independent failures and common failures are required for each subsystem as indicated in Table 2. Since redundancies are considered in the model, values for the beta modification factor in MooN architecture (C_{MooN}) are required. Four redundancies are considered on the 1ooN architecture, Table 3. The model considers the lifecycle cost through the inclusion of purchase, installation, repair, preventive maintenance, test interval maintenance and design cost, Table 4.

In addition to the costs associated to SIS design and installation, other parameters are required to perform a holistic analysis. The data includes the estimation for the plant life considered as 10 years, one year for the test interval, repairing time of 8 h, catastrophic loss of $\$50 \times 10^6$, production layoff loss of $1000/h, start-up cost of $2000, shutdown time of 24 h. To estimate the annual present value, a discount rate Ra = 0.05 was assumed and the risk tolerance criterion considered was 1×10^{-5}. A quench system is incorporated as an additional protection layer as suggested in (Wei et al., 2008) assuming a cost of $200,000 usd.

6. Results

According to the risk tolerance criteria, a SIL 2 is necessary to be implemented in the system. The resulting MINLP model is solved on the General Algebraic Modeling System package (GAMS) using the DICOPT solver. Since the solution is in fact a combination of several devices, the problem is combinatorial and the number of possible configurations for the system in the case study becomes 244,000,000. The optimal configuration of the SIS has a PFD estimated of 9.93×10^{-3} and the STR is 0.142. These results correspond to the technology selection described as follows: A transmitter of technology 4 with an estimated PFD of1.1×10^{-2} is selected for the pressure subsystem, the option 2 for the technology with an estimated PFD of 2.3×10^{-2} is selected for the temperature transmitter, the option 3 for the technology with estimated PFD of 6×10^{-3} is selected for the logic solver, and a technology of type 5 is chosen for the actuator with an estimated PFD of 4×10^{-3}. Thus, a SIL 2 is

Table 2
Independent and common failures.

Device	λ^{SD}	λ^{SU}	λ^{DD}	λ^{DU}	λ^{Total}	β^{SU}	β^{SD}	β^{DU}	β^{DD}	SFF
1 Pressure	8.14	5.2	7.76	5.1	26.2	0.1	0.05	0.1	0.1	0.805
2 Pressure	7.68	10.1	6.84	3.4	28.02	0.1	0.1	0.1	0.1	0.879
3 Pressure	2.34	4.32	6.52	3.21	16.39	0.1	0.1	0.1	0.1	0.804
4 Pressure	2.1	2.74	0.75	2.43	8.02	0.1	0.05	0.1	0.1	0.697
5 Pressure	2.76	1.22	3.4	1.89	9.27	0.1	0.1	0.1	0.1	0.796
1 Temperature	9.84	4.66	5.32	8.33	28.15	0.1	0.02	0.1	0.1	0.805
2 Temperature	3.65	3.34	3.46	5.23	15.68	0.1	0.05	0.1	0.1	0.879
3 Temperature	7.43	5.56	1.54	3.44	17.97	0.1	0.1	0.1	0.1	0.804
4 Temperature	1.13	1.86	1.23	2.35	6.57	0.1	0.02	0.1	0.1	0.697
5 Temperature	0.5	1.36	3.22	1.23	6.31	0.1	0.05	0.1	0.1	0.796
1 PLC	5.64	3.86	2.78	5.5	17.78	0.1	0.1	0.1	0.1	0.805
2 PLC	8.1	3.34	0	2.56	14	0.1	0.1	0.1	0.1	0.879
3 PLC	0.84	1.35	0.4	1.35	3.94	0.1	0.1	0.1	0.1	0.804
4 PLC	1.3056	1.1875	1.2335	1.081	4.8076	0.1	0.1	0.1	0.1	0.697
5 PLC	1.2234	1.36	1.501	0.85	4.9344	0.1	0.1	0.1	0.1	0.796
1 Final item	4.2	4.3	1.34	4.23	14.07	0.1	0.1	0.1	0.1	0.805
2 Final item	1.234	4.01	0.54	3.54	9.324	0.1	0.1	0.1	0.1	0.879
3 Final item	2.44	2.43	1.2	2.65	8.72	0.1	0.1	0.1	0.1	0.804
4 Final item	1.65	1.66	1.6	2.221	7.131	0.1	0.1	0.1	0.1	0.697
5 Final item	1.5443	0.7	1.76	0.86	4.8643	0.1	0.1	0.1	0.1	0.796

Table 3
Values for C_{MooN} redundancies.

Number of devices	C_{MooN}
1	0.0
2	1.0
3	0.3
4	0.15

Table 4
Design, purchase, installation, maintenance, repair and testing costs.

Device	Purchase	Design	Installation	Maintenance	Repair	Testing cost
1 Pressure	35,000	900	430	4200	4200	100
2 Pressure	50,000	900	440	4400	4400	75
3 Pressure	53,000	900	450	4320	4320	85
4 Pressure	58,000	900	450	4500	4500	80
5 Pressure	65,000	900	460	4350	4350	80
1 Temperature	32,000	900	430	4400	4400	120
2 Temperature	40,000	900	440	4400	4400	100
3 Temperature	50,000	900	450	4450	4450	89
4 Temperature	59,000	900	450	4600	4600	65
5 Temperature	68,000	900	460	4300	4300	90
1 PLC	100,000	1200	770	3500	10,050	250
2 PLC	120,000	1200	770	3700	12,050	130
3 PLC	130,000	1200	770	3500	13,050	140
4 PLC	138,000	1200	800	3750	13,850	100
5 PLC	145,000	1200	800	3500	14,550	100
1 Final item	35,000	900	430	4000	3550	120
2 Final item	38,000	900	440	4300	3850	100
3 Final item	41,500	900	450	4200	4200	89
4 Final item	45,000	900	450	3900	4550	65
5 Final item	55,000	900	460	3900	5550	90

Table 5
Final SIS costs.

Concept	Costs ($)
SIS	497,320
Purchase	291,020
Operation	18,317
Corrective maintenance	1607
Preventive maintenance	16,300
Testing	410
Risk	8399
CSTR	3419
Hazard	4980
Total	**767,320**

guarantee for an FT of 0 and an SFF between 60 and 90%. Table 5 shows the resulting costs for the case study.

7. Conclusions

This work has proposed a mathematical approach that aims to select the optimal allocation of IPL's. The proposed model includes initial fixed costs, annual costs such as operation and risk costs in the lifecycle costs of the IPL's to be placed. The optimal design of the SIS has incorporated practical factors such as specific architecture, hardware type, multiple failure modes, diagnostic coverage, common cause failure, proof test intervals, repair times, fault tolerance and safe failure fraction. The proposed MINLP model could be extended to incorporate all other independent protection layers in the optimal LOPA-based design.

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Nomenclature

C_T	total cost ($)
C_T^{Fised}	total initial fixed costs ($)
C_T^{Op}	total annual operation cost ($/year)
C_T^{Risk}	total annual risk cost ($/year)
PV_A	present value of an annuity factor
L	independent protection layer subscript
C_L^{Fised}	initial fixed cost of each independent protection layer
C_L^{Op}	annual operation cost of each independent protection layer ($/year)
R	discount rate
t	lifecycle of the project (years)
C^{RISK}	annual risk cost ($/year)
C^{STR}	annual costs generated by spurious trips ($/year)
C^{HAZARD}	annual catastrophic loss cost due to a dangerous failure ($/year)
CSD	spurious shutdown cost ($)
STR	spurious trip rate
SD_{Loss}	production loss ($)
SD_{Time}	loss production time
C^C	economic catastrophic loss ($)
f^I	initial event frequency
$PFD^{NON-SIS}$	probability of failure on demand of non-SIS independent protection layers
PFD^{SIS}	probability of failure on demand of safety instrumented systems
f^{TR}	risk tolerance criteria (event/year)
C^{SIS}	lifecycle cost of the safety instrumented system ($)
C^{Fixed}	initial fixed costs ($)
C^{DPIC}	design, purchase, installation and commissioning costs ($)
$C^{START-UP}$	Start-up costs ($)
i	type of technology subscript
j	subsystem subscript
r	redundancies subscript
C^T	annual testing costs ($/year)
C_{ij}^T	annual testing costs ($/year) of the element with the i-type of technology of the j-subsystem
C_j^T	annual testing costs ($/year) of the j-subsystem
C^{Op}	annual operation costs ($/year)
C_{ij}^{Op}	annual operation costs ($/year) of the element with the i-type of technology of the j-subsystem
C_j^{Op}	annual operation costs ($/year) of the j-subsystem
C^{CM}	annual corrective maintenance costs ($/year)
C_{ij}^{CM}	annual corrective maintenance costs ($/year) of the element with the i-type of technology of the j-subsystem
C_j^{CM}	annual corrective maintenance costs ($/year) of the j-subsystem
C^{PM}	annual preventive maintenance costs ($/year)
C_{ij}^{PM}	annual preventive maintenance costs ($/year) of the element with the i-type of technology of the j-subsystem
C_j^{PM}	annual preventive maintenance costs ($/year) of the j-subsystem
B_{ijr}	binary variable that corresponds to the number of r redundancies selected, with the i-type of technology of the j-subsystem
M_{ij}	maintenance frequency of the element with the i-type of technology of the j-subsystem
N_r	number of redundancies
f_{ij}	frequency of repair (event/year) of the element with the i-type of technology of the j-subsystem
PDF_{avg}^{SIS}	average probability of failure on demand for a safety function
PDF_{ijr}	probability of failure on demand for the system with r redundancies selected, the i-type of technology of the j-subsystem

PDF_j	probability of failure on demand of the *j*-subsystem
λ_{ij}^{DU}	dangerous undetected failure rate (failures/year) of the element with the *i*-type of technology of the *j*-subsystem
λ_{ij}^{DD}	dangerous detected failure rate (failures/year) of the element with the *i*-type of technology of the *j*-subsystem
λ_{ij}^{SU}	safe undetected failure rate (failure/year) of the element with the *i*-type of technology of the *j*-subsystem
λ_{ij}^{SD}	safe detected failure rate (failures/year) of the element with the *i*-type of technology of the *j*-subsystem
C_{MooN}	configuration factor
β_{ij}^{DU}, β_{ij}^{DD}	parameters corresponding to the β factor model for dangerous detected and undetected failures of the element with the *i*-type of technology of the *j*-subsystem
β_{ij}^{SU}, β_{ij}^{SD}	parameters corresponding to the β factor model for safe detected and undetected failures of the element with the *i*-type of technology of the *j*-subsystem
Tr	repair time (h)
TI	test interval (years)

Acronyms

LOPA	layer of protection analysis
IPL	independent protection layer
SIL	safety integrity level
SIS	safety instrumented systems

References

Andrews, J. D., & Ericson, C. A., II. (2000). Fault tree and Markov analysis applied to various design complexities. In *18th International system safety conference*, Texas, USA.

CCPS. (2001). *Layer of protection analysis (LOPA): Simplified process risk analysis*. New York: CCPS/AIChE.

Dowell, A. M. (1998). Layer of protection analysis for determining safety integrity level. *ISA Transactions, 37*(3), 155–165.

Dowell, A. M. (1999). Layer of protection analysis and inherently safer processes. *Process Safety Progress, 18*(4), 214–220.

Goble, W. M. (2010). *Control systems safety evaluation and reliability*. International Society of Automation.

Goble, W. M., & Cheddie, H. (2005). *Safety instrumented systems verification*. Research Triangle Park, NC, US: S. a. A. S. The Instrumentation.

IEC 61508. (1998–2005). *Functional safety of electrical/electronic/programmable electronic safety-related systems*. Parts 1–7. Switzerland: International Electrotechnical Commission.

Markowski, A. S., & Mannan, M. S. (2009). Fuzzy logic for piping risk assessment (pfLOPA). *Journal of Loss Prevention in the Process Industries, 22*(6), 921–927.

Markowski, A. S., & Mannan, M. S. (2010). ExSys–LOPA for the chemical process industry. *Journal of Loss Prevention in the Process Industries, 23*, 688–696.

Pasman, H. J. (2000). Risk informed resource allocation policy: safety can save costs. *Journal of Hazardous Materials, 71*, 375–394.

Schupp, B., Hale, A., Pasman, H., Lemkovitz, S., & Goossens, L. (2006). Design support for the systematic integration of risk reduction into early chemical process design. *Safety Science, 44*(1), 37–54.

Smith, D. J., & Simpson, K. G. (2011). *A straightforward guide to functional safety: IEC 61508 (2010 edition) and related standards, including: Process IEC 61511, machinery IEC 62061 and ISO 13849. Safety critical systems handbook* (3rd ed.).

Stauffer, T. (2010). Making the most of alarms as a safety layer of protection. In *Safety systems conference*, Calgary, AB, Canada.

Summers, A. E. (2003). Introduction to layers of protection analysis. *Journal of Hazardous Materials, 104*, 163–168.

Tixier, J., Dusserre, G., Salvi, O., & Gaston, D. (2002). Review of 62 risk analysis methodologies of industrial plants. *Journal of Loss Prevention in the Process Industries, 15*(4), 291–303.

Torres-Echeverria, A. C. (2009a). *Modelling and optimization of safety instrumented systems based on dependability and cost measures*. Doctor of Philosophy, Department of Automatic Control and Systems Engineering, University of Sheffield.

Torres-Echeverria, A. C. (2009b). *Modelling and optimization of safety instrumented systems based on dependability and cost measures* (p. 301). PhD, Department of Automatic Control and Systems Engineering, University of Sheffield.

Wei, C., Rogers, W. J., & Mannan, M. S. (2008). Layer of protection analysis for reactive chemical risk assessment. *Journal of Hazardous Materials, 159*(1), 19–24.

Yun, G., Rogers, W. J., & Mannan, M. S. (2009). Risk assessment of LNG importation terminals using the Bayesian-LOPA methodology. *Journal of Loss Prevention in the Process Industries, 22*(1), 91–96.